第四章 (24, 26, 45, 46, 47, 50, 53, 55, 82)

24. (a) $p-(1-p)\frac{3}{4}=\frac{7}{4}p-3/4$, (b) $-\frac{3}{4}p+(1-p)2=-\frac{11}{4}p+2$

$\frac{7}{4}p-3/4=-\frac{11}{4}p+2 \Rightarrow p=11/18$, maximum value = 23.72

(c) $q-\frac{3}{4}(1-q)$ (d) $-\frac{3}{4}q+2(1-q)$, minimax value = 23/72

attained when q = 11/18

26.

26、令 X 為問問題數目

(a) $E[X] = 1\times\frac{1}{10}+2\times\frac{1}{10}+\cdots\cdots+10\times\frac{1}{10}=\frac{55}{10}=\frac{11}{2}$

(b) 每次問題要從剩下的數字中從可能去掉近半之數字。

$\Rightarrow$ (1st) Q：是不是在 1~5 $\begin{cases}\text{是} \Rightarrow \text{剩下 } 1\sim5 \text{ —①}\\ \text{不是} \Rightarrow \text{剩下 } 6\sim10 \text{ —②}\end{cases}$

若① $\Rightarrow$ (2nd) Q：是不是在 1~3 $\begin{cases}\text{是} \Rightarrow \text{剩下 } 1\sim3\\ \text{不是} \Rightarrow \text{剩下 } 4\sim5\end{cases}$

若② $\Rightarrow$ (2nd) Q：是不是在 6~8 $\begin{cases}\text{是} \Rightarrow \text{剩下 } 6\sim8\\ \text{不是} \Rightarrow \text{剩下 } 9\sim10\end{cases}$

$\therefore$ 問完2次後剩下3個數的機率為 $\frac{3}{5}$, $\frac{2}{5}$ 下一頁

$\therefore E[X] = 3\cdot\frac{3}{5}+2\cdot\frac{2}{5}$

2+1 問完2次剩下2個數，只須多問1次

$+\frac{3}{5}\left[3\times\frac{1}{3}+4\times\frac{2}{3}\right]$

問完2次剩下3個數

2+1

問第3次

Q：(是不是其中2個 $\Rightarrow$ Ans：不是)

意謂答案在剩下那個

2+2

(3rd) Q：是不是其中2個 $\Rightarrow$ Ans：是

$\therefore$ (4th) Q：是不是2個其中1個

Ans：不管是不是都能找出答案

$= 3\cdot\frac{2}{5}+\frac{3}{5}\left[3\cdot\frac{1}{3}+4\cdot\frac{2}{3}\right]$

$= \frac{6}{5}+\frac{3+8}{5}=\frac{17}{5}$ #

45. $\alpha\sum_{i=k}^{n}\binom{n}{i}p_1^i(1-p_1)^{n-i}+(1-\alpha)\sum_{i=k}^{n}\binom{n}{i}p_2^i(1-p_2)^{n-i}$

46. with 3: $P\{\text{pass}\} = \frac{1}{3}\left[\binom{3}{2}(.8)^2(.2)+(.8)^3\right]+\frac{2}{3}\left[\binom{3}{2}(.4)^2(.6)+(.4)^3\right]$

$= .533$

with 5: $P\{\text{pass}\} = \frac{1}{3}\sum_{i=3}^{5}\binom{5}{i}(.8)^i(.2)^{5-i}+\frac{2}{3}\sum_{i=3}^{5}\binom{5}{i}(.4)^i(.6)^{5-i}$

$= .3038$

47. Let C be the event that the jury is correct, and let G be the event that the defendant is guilty.

Then

$$\begin{aligned}P(C) &= P(C|G)P(G)+P(C|G^c)P(G^c)\\ &= \sum_{i=0}^{3}(.2)^i(.8)^{12-i}(.65)+\sum_{i=0}^{8}(.1)^i(.9)^{12-i)}(.35)\end{aligned}$$

Let CV be the event that the defendant is convicted. Then

$$\begin{aligned}P(CV) &= P(CV|G)P(G)+P(CV|G^c)P(G^c)\\ &= \sum_{i=0}^{3}(.2)^i(.8)^{12-i}(.65)+\left(1-\sum_{i=0}^{8}(.1)^i(.9)^{12-i)}(.35)\right)\\ &= \sum_{i=0}^{3}(.2)^i(.8)^{12-i}(.65)+\sum_{i=9}^{12}(.1)^i(.9)^{12-i)}(.35)\end{aligned}$$

50. (a) $\frac{1}{2}\binom{10}{7}.4^7.6^3+\frac{1}{2}\binom{10}{7}.7^7.3^3$

(b) $\dfrac{\frac{1}{2}\binom{9}{6}(.4)^7(.6)^3+\frac{1}{2}\binom{9}{6}(.7)^7(.3)^3}{.55}$

53. $P(Win, Win, Lose|7\ Wins) = \frac{P(Win, Win, Lose\ and\ 7\ Wins)}{P(7\ Wins)} =$

$$\frac{P(7\ Wins|Win,Win,Lose)P(Win,Win,Lose)}{P(7\ Wins)} = \frac{P(5\ Wins\ in\ 7\ Rounds)P(Win,Win,Lose)}{P(7\ Wins)} =$$

$$\frac{\left(\binom{7}{5}p^5(1-p)^2\right)\left(p^2(1-p)^1\right)}{\binom{10}{3}p^7(1-p)^3} = \frac{7}{40}$$

(a)

$$P(Lose, Win, Lose|7\ Wins) = \frac{P(Lose,Win,Lose\ and\ 7\ Wins)}{P(7\ Wins)} =$$

$$\frac{P(7\ Wins|Lose,Win,Lose)P(Lose,Win,Lose)}{P(7\ Wins)} = \frac{P(6\ Wins\ in\ 7\ Rounds)P(Lose,Win,Lose)}{P(7\ Wins)} =$$

$$\frac{\left(\binom{7}{6}p^6(1-p)^1\right)\left(p^1(1-p)^2\right)}{\binom{10}{3}p^7(1-p)^3} = \frac{7}{120}$$

55. (a) $1 - e^{-3.5} - 3.5e^{-3.5} = 1 - 4.5e^{-3.5}$

(b) $4.5e^{-3.5}$

Since each flight has a small probability of crashing it seems reasonable to suppose that the number of crashes is approximately Poisson distributed.

82. Let F be the number of students in the group who have failed the test.

(a) $P(F = 0) = \frac{\binom{41}{10}}{\binom{50}{10}} \approx 0.1091$

(b) $P(F \geq 3) = 1 - P(F \leq 2) = 1 - \frac{\binom{41}{10}+\binom{41}{9}\binom{9}{1}+\binom{41}{8}\binom{9}{2}}{\binom{50}{10}} \approx 0.2491$

第五章 (22, 23, 32, 35, 36, 37, 38, 40)

22. Let S_n denote the number of times Tom hits the bullseye in n throws and Z a standard normal random variable. Then,

$$P(S_{50} < 1) = P(S_{50} < 0.5) = P\left(\frac{S_{50} - (50)(0.05)}{\sqrt{(50)(0.05)(0.95)}} < \frac{0.5 - (50)(0.05)}{\sqrt{(50)(0.05)(0.95)}}\right)$$

$$\approx P\left(Z < \frac{0.5 - (50)(0.05)}{\sqrt{(50)(0.05)(0.95)}}\right) = \Phi(1.6222) = 0.9476.$$

23. (a) Let R_n denote the number of times a red card appears in n picks and Z be a standard normal random variable. Then,

$P(249.5 < R_{100} < 300.5) = P\left(\frac{249.5-500(0.5)}{\sqrt{500(0.5)(0.5)}} < \frac{R_{100}-500(0.5)}{\sqrt{500(0.5)(0.5)}} < \frac{300.5-500(0.5)}{\sqrt{500(0.5)(0.5)}}\right) \approx$

$P\left(\frac{249.5-500(0.5)}{\sqrt{500(0.5)(0.5)}} < Z < \frac{300.5-500(0.5)}{\sqrt{500(0.5)(0.5)}}\right) = P(-0.0447 < Z < 4.5169) =$

$\Phi(4.5169) - \Phi(-0.0447) = 0.5178.$

(b) Let E_n denote the number of times an even card appears in n picks and Z be a standard normal random variable. Then,

$P(E_{100} > 200) = P\left(\frac{E_{100}-500\left(\frac{20}{52}\right)}{\sqrt{500\left(\frac{20}{52}\right)\left(\frac{32}{52}\right)}} > \frac{200-500\left(\frac{20}{52}\right)}{\sqrt{500\left(\frac{20}{52}\right)\left(\frac{32}{52}\right)}}\right) \approx P\left(Z > \frac{200-500\left(\frac{20}{52}\right)}{\sqrt{500\left(\frac{20}{52}\right)\left(\frac{32}{52}\right)}}\right) =$

$P(Z > 0.7071) = 1 - \Phi(0.7071) = 0.2398.$

32.

$\frac{1}{\lambda} = 1.5 \Rightarrow \lambda = \frac{2}{3}$; therefore, $P(X > x) = 1 - F(x) = e^{-\frac{2}{3}x}$.

(a) $P(X > 2) = e^{-\frac{4}{3}} = 0.2636.$

(b) By memoryless property, $P(X > 2 \mid X > 1) = P(X > 1) = e^{-\frac{2}{3}} = 0.5134.$

35. $F_Y(y) = P(Y \le y) = P\left(\sqrt{X} \le y\right) = P(X \le y^2) = 1 - e^{-\lambda y^2}$ for $y \ge 0$.
Therefore, the probability density function is given

$$f_Y(y) = \frac{dF_Y(y)}{dy} = \begin{cases} 2\lambda y e^{-\lambda y^2}, & y \ge 0 \\ 0, & otherwise \end{cases}$$

Y is a Weibull random variable with parameters $v = 0, \alpha = 1, and\ \beta = 2.$

36. (a) $exp\left[-\int_5^{10} \frac{1}{2t} dt\right] = 0.7071.$

(b) $exp\left[-\int_5^{25} \frac{1}{2t} dt\right] = 0.4472.$

37. (a) $exp\left[-\int_0^4 (0.25 + 0.14t^2)dt\right] = 0.0186.$

(b) $exp\left[-\int_0^{2.3} (0.25 + 0.14t^2)dt\right] - exp\left[-\int_0^{3.7} (0.25 + 0.14t^2)dt\right] = 0.3189 -$

$0.0373 = 0.2816.$

(c) $exp\left[-\int_2^5 (0.25+0.14t^2)dt\right] = 0.0020$.

38. (a) $F_Y(y) = P(Y \le y) = P(log(X) \le y) = P(X \le e^y) = \frac{e^y-1}{4}$ for $log(1) \le y \le log(5)$. Therefore, the probability density function is given by the following:

$$f_Y(y) = \frac{dF_Y(y)}{dy} = \begin{cases} \frac{1}{4}e^y, & log(1) \le y \le log(5) \\ 0, & otherwise \end{cases}$$

(b) $P\left(\frac{1}{3} < Y < \frac{2}{3}\right) = F_Y\left(\frac{2}{3}\right) - F_Y\left(\frac{1}{3}\right) = 0.1380$.

40. $F_Y(y) = P(Y \le y) = P\left(e^{-\lambda X} \le y\right) = P\left(log\left(e^{-\lambda X}\right) \le log(y)\right) = P\left(X \ge -\frac{1}{\lambda}log(y)\right) = e^{-\lambda\left(-\frac{1}{\lambda}log(y)\right)} = y$ for $0 < y \le 1$.

Therefore, the probability density function is given by the following equation:

$$f_Y(y) = \frac{dF_Y(y)}{dy} = \begin{cases} 1, & y \in (0,1] \\ 0, & otherwise, \end{cases}$$

, i.e., Y is a uniform random variable on (0,1].

第六章 (8, 9, 15, 19, 21, 22, 23)

8. Because $\int_0^1\int_0^1 c(x+y^3)dxdy = c\int_0^1 \frac{1}{2}+y^3 dy = \frac{3}{4}c = 1$, $c = 4/3$.

(a) The marginal density of X is $f(x) = \int_0^1 \frac{4}{3}(x+y^3)dy = \frac{1}{3}(4x+1)$, $0 \le x \le 1$ and the marginal density of Y is $f(y) = \int_0^1 \frac{4}{3}(x+y^3)dx = \frac{2}{3}(1+2y^3)$, $0 \le y \le 1$.

(b) $E[XY] = \frac{4}{3}\int_0^1\int_0^1 xy(x+y^3)dxdy = \frac{4}{3}\int_0^1 \frac{y}{3}+\frac{y^4}{2}dy = \frac{16}{45}$.

9. Because $c\int_0^1\int_0^x xe^{-y}dydx = c\int_0^1 x(1-e^{-x})dx = \frac{e^{-1}(4-e)}{2} = 1$, $c = \frac{2}{e^{-1}(4-e)}$.

(a) From (a) we deduce that $f(x) = \frac{2x(1-e^{-x})}{e^{-1}(4-e)}$ for $0 \le x \le 1$.

(b) $f(y) = \int_y^1 \frac{2xe^{-y}}{e^{-1}(4-e)}dx = \frac{(1-y^2)e^{-y}}{e^{-1}(4-e)}$ for $0 \le y \le 1$.

(c) $P\left(Y < \frac{1}{3} \middle| X < \frac{1}{2}\right) = \frac{P\left(Y<\frac{1}{3} and\ X<\frac{1}{2}\right)}{P\left(X<\frac{1}{2}\right)} = \frac{\int_0^{\frac{1}{3}}\int_0^x xe^{-y}dydx + \int_{\frac{1}{3}}^{\frac{1}{2}}\int_0^{\frac{1}{3}} xe^{-y}dydx}{\int_0^{\frac{1}{2}} x(1-e^{-x})dx} \approx 0.8799.$

(d) $E[X] = \int_0^1 \frac{2x^2(1-e^{-x})}{e^{-1}(4-e)}dx \approx 0.7327.$

(e) $E[Y] = \int_0^1 \frac{y(1-y^2)e^{-y}}{e^{-1}(4-e)}dy \approx 0.3188.$

15. (a) $1 = \iint f(x,y)dydx = \int\limits_{(x,y)\in R}\int c\, dydx = cA(R)$

where $A(R)$ is the area of the region R.

(b) $f(x, y) = 1/4, -1 \le x, y \le 1$

$= f(x)f(y)$

where $f(v) = 1/2, -1 \le v \le 1$.

(c) $P\{X^2 + Y^2 \le 1\} = \frac{1}{4}\iint\limits_C dydx$ = (area of circle)/4 = $\pi/4$.

19. $f(x,y) = \frac{x(1-x)}{B(3,2)}, 0 < y < x < 1$ is a density function because $\int_0^1\int_0^x x(1-x)dydx = \int_0^1 x^2(1-x) = B(3,2)$; therefore,

(a) the marginal density of X is $f(x) = \frac{x^2(1-x)}{B(3,2)}, 0 < x < 1$;

(b) the marginal density of Y is $f(y) = \frac{1}{B(3,2)}\int_y^1 x(1-x)dx = \frac{2y^3-3y^2+1}{6B(3,2)}, 0 < y < 1$;

(c) $E[X] = \int_0^1 \frac{x^3(1-x)}{B(3,2)}dx = \frac{3}{5}$; and

(d) $E[Y] = \int_0^1 y\frac{2y^3-3y^2+1}{6B(3,2)}dy = 0.3.$

21. $\int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \frac{1}{2} \sin(x+y)\, dxdy = \frac{1}{2} \int_0^{\frac{\pi}{2}} \sin(y) + \cos(y)\, dy = 1.$

(a) $E[X] = \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \frac{1}{2} x \sin(x+y)\, dxdy = \frac{\pi}{4}.$

(b) $E[\cos Y] = \int_0^{\frac{\pi}{2}} \int_0^{\frac{\pi}{2}} \frac{1}{2} \cos y \sin(x+y)\, dxdy = \frac{\pi+2}{8}.$

22. (a) No, since the joint density does not factor.

(b) $f_X(x) = \int_0^1 (x+y)dy = x + 1/2, \quad 0 < x < 1.$

(c) $P\{X + Y < 1\} = \int_0^1 \int_0^{1-x} (x+y)dydx$

$$= \int_0^1 [x(1-x) + (1-x)^2 / 2]dx = 1/3$$

23. (a) yes

$$f_X(x) = 12x(1-x)\int_0^1 ydy = 6x(1-x), \quad 0 < x < 1$$

$$f_Y(y) = 12y \int_0^1 x(1-x)dx = 2y, 0 < y < 1$$

(b) $E[X] = \int_0^1 6x^2(1-x)dx = 1/2$

(c) $E[Y] = \int_0^1 2y^2 dy = 2/3$

(d) $\mathrm{Var}(X) = \int_0^1 6x^3(1-x)dx - 1/4 = 1/20$

(e) Var(Y) = $\int_0^1 2y^3 dy - 4/9$ = 1/18

第八章 (7, 8、self-test 1, 2, 4, 7)

7. $$P\left\{\sum_{i=1}^{100} X_i > 525\right\} \approx P\left\{N(0,1) > \frac{525-500}{\sqrt{(100\times 25)}}\right\} = P\{N(0,1) > .5\} = .3085$$

where the above uses that an exponential with mean 5 has variance 25.

8. If we let X_i denote the life of bulb i and let R_i be the time to replace bulb i then the desired probability is $P\left\{\sum_{i=1}^{100} X_i + \sum_{i=1}^{99} R_i \le 550\right\}$. Since $\sum X_i + \sum R_i$ has mean $100 \times 5 + 99 \times .25 = 524.75$ and variance $2500 + 99/48 = 2502$ it follows that the desired probability is approximately equal to $P\{N(0, 1) \le [550 - 524.75]/(2502)^{1/2}\} = P\{N(0, 1) \le .505\} = .693$

It should be noted that the above used that

$$\text{Var}(R_i) = \text{Var}\left(\frac{1}{2}\text{Unif}[0,1]\right) = 1/48$$

self-test

1. 令 X 表下週之銷售數量，注意 X 是整數的。由馬可夫不等式，我們得到下列：

(a) $P\{X > 18\} = P\{X \ge 19\} \le \frac{E[X]}{19} = 16/19$

(b) $P\{X > 25\} = P\{X \ge 26\} \le \frac{E[X]}{26} = 16/26$

2. (b)小題解答分母打錯，應為 9 + 9

2. (a) $P\{10 \le X \le 22\} = P\{|X-16| \le 6\}$

$$= P\{|X-\mu| \le 6\}$$
$$= 1 - P\{|X-\mu| > 6\}$$
$$\ge 1 - 9/36 = 3/4$$

(b) $P\{X \ge 19\} = P\{X-16 \ge 3\} \le \dfrac{9}{9-9} = 1/2$

在 (a) 部分，我們使用謝比雪夫不等式；在 (b) 部分，我們使用其單邊型式。(見定理 5.1)

4. 若 X 為工廠 A 所生產之數量且 Y 為工廠 B 所生產之數量，則

$$E[Y-X] = -2, \quad \mathrm{Var}(Y-X) = 36+9 = 45$$

$$P\{Y-X>0\} = P\{Y-X \ge 1\} = P\{Y-X+2 \ge 3\} \le \frac{45}{45+9} = 45/54$$

7. 若 X 表維修一部機器所需時間，則

$$E[X] = 0.2 + 0.3 = 0.5$$

而且，指數隨機變數的變異數等於其平均數的平方，所以

$$\mathrm{Var}(X) = (0.2)^2 + (0.3)^2 = 0.13$$

因此，令 X_i 表服務工作 i 所花時間 $i = 1, \cdots, 20$，且 Z 為一個標準常態隨機變數，則

$$P\{X_1 + \cdots + X_{20} < 8\} = P\left\{\frac{X_1 + \cdots + X_{20} - 10}{\sqrt{2.6}} < \frac{8-10}{\sqrt{2.6}}\right\}$$
$$\approx P\{Z < -1.24035\}$$
$$\approx .1074$$